MILITARY SPECIFICATIONS

CARTRIDGE, DELAY: M293

1. SCOPE

1.1 This specification covers Cartridge, Delay, M293 for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL—A—2550	Ammunition and Special Weapons, General Specification for.
MIL—E—5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL—C—60539	Coating, Anodic, Conventional, for Aluminum Aluminum Alloys.

STANDARDS

Military

MIL—STD—105	Sampling Procedures and Tables for Inspection by Attributes.
MIL—STD—109	Quality Assurance Terms and Definitions.
MIL—STD—413	Visual Inspection Guide for Rubber 0 Rings.
MS—28741	Hose Assembly, Detachable End Fitting, Medium Pressure.

DRAWINGS

Munitions Command

D11730546	Cartridge, Delay: M293.
C8799621	Carton, Packing, Ammunition for Cartridge, Delay, M70, M93, M113, M155 or M293.
C8799620	Box, Packing, Ammunition, for Cartridge, Delay, M70, M93, M113, M155, or M293.
IEL11730546	Inspection Equipment List for Cartridge Delay, M293.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. the following documents forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply:

ASA B46.1—Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N. Y. 10016.)

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3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish. Protective finish shall be in accordance with MIL—C—60539.

3.3 Vibration. The cartridge shall withstand vibration as prescribed in MIL—E—5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant or derangement of any components.

3.4 Functional.

3.4.1 The cartridge shall not fail to fire.

3.4.2 *Ballistics.* The cartridge when tested in an XM-120 Initiator shall produce the pressures and delay times at the temperatures specified in Table I. Pressure values shall be recorded at the end of 15 feet of aircraft hose, Size No. 4, in accordance with MS28741 and a pressure chamber with a volume of 1 cubic inch:

TABLE I

Temperature	Pressure Pounds Per Square Inch (PSI)		Delay Time (Seconds) (See 6.2)
	Min	Max	
-65° to -70° F	500	1250	.780 Max.
65° to 75° F	600	1500	.650 ±.060
200° to 205° F	700	1700	.520 Min.

3.5 Workmanship. Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.5.1 *Metal defects.* The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.5.2 *Burr.* The metal part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 *Foreign matter.* The metal part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality Assurance Terms and Definitions.* Reference shall be made to MIL—STD—109 to define quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with MIL—A—2550.

4.2 First Article Sample.

4.2.1 *Initial production sample.* At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 50 uncoated cartridge cases for hardness tests, 23 coated cases for protective finish tests, 5 cartridges for sealant tests and 30 cartridges to be tested functionally as indicated in Table II herein, and 10 sets of unassembled components (including explosive components) for visual and dimensional examination.

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TABLE II

Paragraphs		Test	Sample Size	Type of Test	
Requirement	Test			Destructive	Non-Destr.
Dwg.	4.4.1	Hardness	50 (uncoated)	X	—
3.2	4.4.2	Pro. Finish	23 ca	X	—
3.3	4.4.3	Vibration	12 (From Ballistics)	—	X
Dwg.	4.4.4	Sealant app.	5 ea.	X	—
Dwg.	4.4.5	Airtightness	Each	—	X
3.4	4.4.6	Ballistics	30 Ea.	X	—

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 *Examination and tests.* After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 *Ballistics.* The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with Table I. The sample for firing shall consist of the following:

TABLE III

Temperature	Not Vibrated	Vibrated
-65° to -70° F	6	4
65° to 75° F	6	4
200° to 205° F	6	4

4.2.1.3 *Initial production sample failure.* Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 *Lot.*

4.3.1.1 *Submission of products.* The product shall be submitted in accordance with MIL—STD—105.

4.3.1.2 *Size of Lot.* An ammunition lot shall consist of an identifiable group of metal parts assemblies consisting of:

(a) Components produced by a homogeneous process.

(b) Delay element from not more than one lot.

(c) Primers from not more than one lot.

(d) Propellant from not more than one lot.

(e) Igniters from not more than one lot.

4.3.2 *Inspection.* One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification or defect using the sampling plan and associated criteria in Table IV and V. For an individual defect (single characteristic) the criteria in Table IV shall be used. For total defects within a major or minor class, the criteria in Table V shall be used. All non-conforming material shall be rejected. Four power (4X) magnification may be used to perform visual examination for such defects as burr and foreign matter. The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination.

4.3.2.1 *Sealing Disc, Drawing C8594077, Covering a detail of Drawing D11730546.*

	Defect	Method of Inspection
CRITICAL		
1.	Metal defective	Visual
2.	Marking Missing, incorrect or illegible	Visual
MAJOR		
101.	Diameter of body, min.	Gage
102.	Thickness	Gage
103.	Protective finish inadequate or defective	Visual
104.	Burr	Visual
105.	Foreign matter	Visual

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TABLE IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1—300	100% Examination				
301—1300	110	1	2	2	3
1301—3200	150	2	3	3	4
3201—8000	225	3	4	4	5
8001—22000	300	4	5	5	6

TABLE V

Sample Size for Major Defects Number of Defects in Class	Total Allowable Defects per Sample Size			
	110	150	225	300
2—3	1	2	3	4
4—7	2	3	4	5
8—19	3	4	5	7
20—43	4	5	8	10
Over 43	6	8	11	14
Sample Size for Minor Defects Number of Defects in Class	**110**	**150**	**225**	**300**
2—3	2	3	4	5
4—7	3	4	5	7
8—16	4	5	8	10
17—48	6	8	11	14
Over 48	8	11	17	20

MINOR

201.	Diameter of body, max.	Gage
202.	Varnish missing or inadequate	Visual

4.3.2.2 *Case, Cartridge, Drawing C8595202, Covering a detail of Drawing. D11730546.*

Defect		Method of Inspection
CRITICAL		
1.	Metal defective	Visual
MAJOR		
101.	Concentricity of head with body	Gage
102.	Straightness of base	Gage
103.	Diameter of body, min.	Gage
104.	Diameter of head, min.	Gage
105.	Thickness of base	Gage
106.	Thickness of wall	Gage
107.	Location of datum diameter	Gage
108.	Protective finish inadequate or defective	Visual
109.	Burr or sharp corner	Visual
110.	Foreign matter	Visual
MINOR		
201.	Diameter of head, Max.	Gage
202.	Diameter of body, Max.	Gage
203.	Total length	Gage
204.	Radius or chamfer missing or incorrect	Visual
205.	Surface finish improper	Visual

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4.3.2.3 *Delay Element, M75, Dwg. C11730545 Covering a detail of Drawing D11730546.*

Defect	Method of Inspection
CRITICAL	
1. Mixed delay elements	Visual
2. Flash hole missing or obstructed	Visual
3. Primer uncrimped or seal missing	Visual
4. Marking missing, incorrect or illegible	Visual
5. Delay charge not visible through flash hole	Visual
6. Metal defective	Visual
MAJOR	
101. Protective finish inadequate or defective	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined.	

4.3.2.4 *Cartridge, Delay, M293, Drawing D11730546.*

Defect	Method of Inspection
CRITICAL	
1. Any part missing	X-ray & Visual 1/
2. Improper assembly	X-ray & Visual 1/
3. Mixed delay element or propellant	Visual
4. Metal defective	Visual
5. Crimp missing or incorrect	Visual
6. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Cartridge profile failure	Gage
102. Diameter of body, max.	Gage
103. Diameter of head, max.	Gage
104. Location of datum diameter	Gage
105. "O" Ring split, cracked or cut	Visual
106. Protective finish inadequate or defective	Visual
107. Foreign matter	Visual
108. Burr	Visual
MINOR	
201. Total length, max.	Gage
202. Case damaged	Visual
203. Varnish missing or inadequate	

1/ The cartridge shall be x-rayed with the long axis of the case perpendicular to the rays. The x-ray negatives properly identified, shall be forwarded to the procuring activity.

4.3.2.5 *Carton, Packing, Ammunition for Cartridge, Delay M293 Dwg. C8799621.*

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose, or distorted	Visual
205. Cut or damage through impregnated box	Visual
206. Marking missing, incorrect, or illegible	Visual

4.3.2.6 *Box, Packing, Ammunition for Cartridge, Delay M-293 Dwg. C8799620.*

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect, or illegible	Visual

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207.	Number of cartons incorrect	Visual
208.	Car seal missing, unscaled or improperly positioned	Visual

4.3.3 *Inspection testing.* The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 *Hardness.* Fifty units from each heat treated lot shall be tested. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.2 *Protective finish.* Twenty-three units of each component shall be selected for protective finish tests. Failure of any component to comply with the requirement of 3.2 shall be cause for rejection of the lot.

4.3.3.3 *Vibration.* Twelve of the assembled cartridges of the initial production lot or 12 assembled cartridges from a subsequent lot when a change is made on the lot of propellant igniter or delay element, shall be subjected to vibration tests. After vibration the 12 cartridges shall be ex-rayed in accordance with note 1/ of 4.3.2.4 for evidence of deterioration of the primer or propellant or derangement of metal parts. Failure of any cartridge to comply with the requirements of 3.3 shall be cause for rejection of the lot.

4.3.3.4 *Sealant application.* Five cartridges from each production lot shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot of cartridges.

4.3.3.5 *Airtightness.* Each cartridge assembly shall be subjected to the specified pressure differential. Failure of any cartridge to comply with the requirements of the drawings shall be cause for rejection of the cartridge.

4.3.3.6 *Functional.* The following test shall be performed by the Government.

4.3.3.6.1 *Ballistics.* The 30 assembled cartridges from each production lot shall be grouped and temperature conditioned in accordance with 4.4.6.1.1. Failure of any cartridge to comply with the requirements of 3.4 shall be cause for rejection of the lot.

4.3.4 *Inspection equipment.* The examination and tests shall be made using equipment listed on IEL—11730546.

4.4 Test method and procedures.

4.4.1 *Hardness.* A hardness testing machine shall be used to determine compliance with drawing requirements.

4.4.2 *Protective finish.* Test procedures for protective finish shall be in accordance with MIL—C—60539. The units shall be sectioned longitudinally to better expose the critical surfaces to the salt spray.

4.4.3 *Vibration.* The method and procedure for performing the vibration test shall be in accordance with MIL—E—5272, Procedure XII.

4.4.4 *Sealant application.* The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of the joining surfaces at the junction of the sealing disc and case sidewall and that the propellant cavity is free of sealant.

4.4.5 *Airtightness.* The method for determining airtightness shall be by use of equipment listed on IEL 11730546.

4.4.6 *Functional.*

4.4.6.1 *Ballistics.* A sample of 30 assembled cartridges from the initial production sample (including the 12 vibrated cartridges) shall be ballistically tested after temperature conditioning in accordance with Table VI and 4.4.6.1.1. This table shall apply if there has been any change in process or components indicated.

4.4.6.1.1 *Temperature conditioning.* Cartridges shall be loaded in acceptable XM120 Initiators with end caps to cover the gas ports during the conditioning period. The temperature of the propellant within the cartridges shall be considered as the conditioning temperature and will be determined by means of a thermocouple attached to the center of the propellant charge of a specimen cartridge assembled in a standard test device similar to the Initiator undergoing tests. The conditioning time shall be 1 hour longer than the time necessary to bring the element contained in the specimen cartridge to the required temperature.

TABLE VI

Temperature	Sample Size
-65°F to -70°F	10
65°F to 75°F	10
200°F to 205°F	10

5. PREPARATION FOR DELIVERY.

5.1 Packing and marking. Packing and marking shall be in accordance with Dwgs. C8799620 and C8799621.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number and date of this specification.

6.2 Delay time. Delay time is defined as the time that elapses between the firing of the primar and the beginning of a continuous pressure rise on a pressure time record.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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